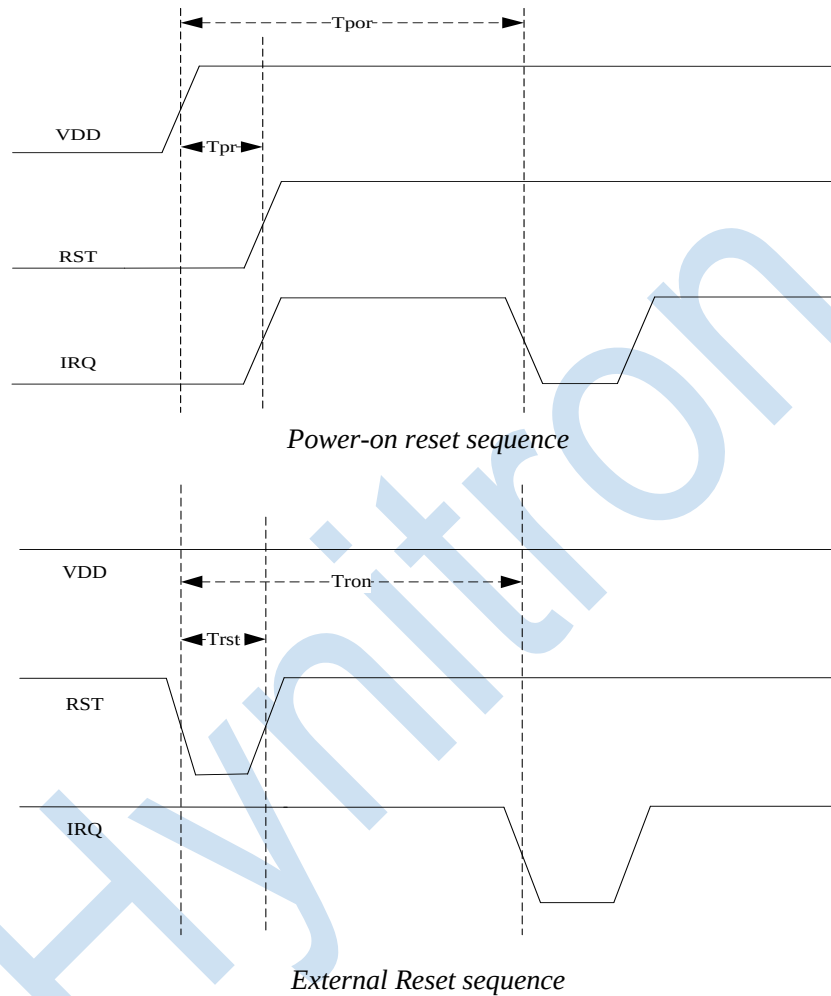


1. Power On and reset

The chip has a built-in power-on reset circuit, so there is no need to connect a dedicated reset circuit externally.

The built-in power-on reset module will keep the chip in the reset state until the voltage is normal. When the voltage is lower than a certain threshold, the chip will also be reset.

When the external reset pin RST is low, the entire chip will be reset, and this pin can be left floating.



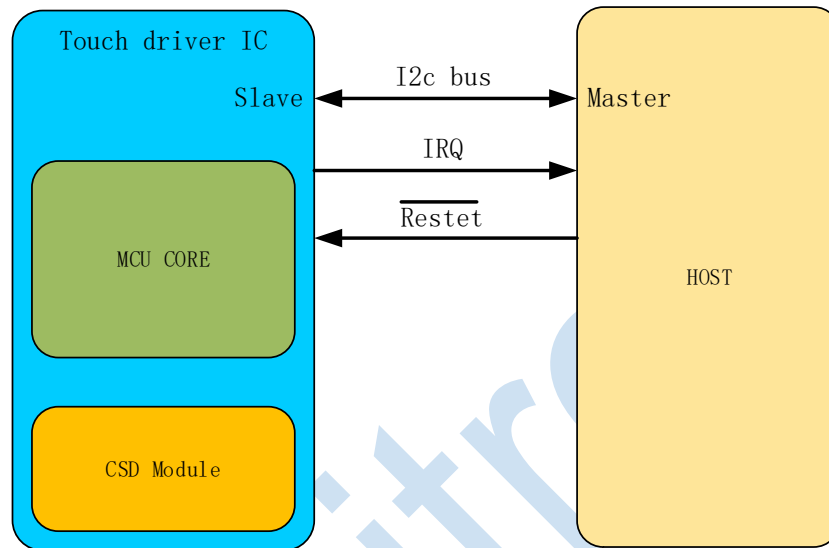
Parameter	Description	minimum	maximum	unit
T_{por}	Chip initialization time after power-on	100	-	mS
T_{pr}	RST pin delayed pull-up time	5	-	mS
T_{ron}	Chip reinitialization time after reset	100	-	mS
T_{rst}	reset pulse time	0.1	-	mS

2. I2C interface description

communication interface

As shown in the figure below, there are the following three types of communication between the host and the touch IC

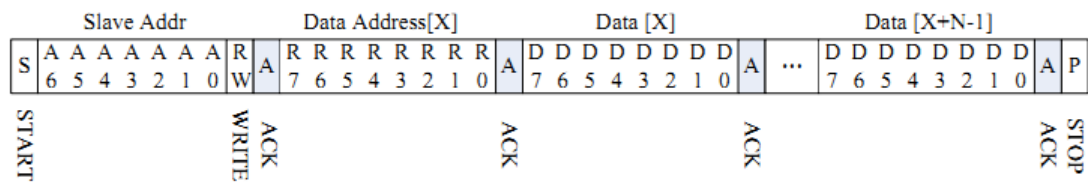
- 1、 IIC communication for data transmission
- 2、 There is touch data report to notify the host through interruption
- 3、 The host resets the touch IC through the Rst pin



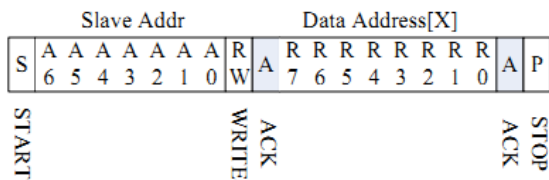
The default slave device address is 0x15 (7bit). The device address of some projects may be different. Please consult the corresponding project and engineering personnel.

3. I2C read and write

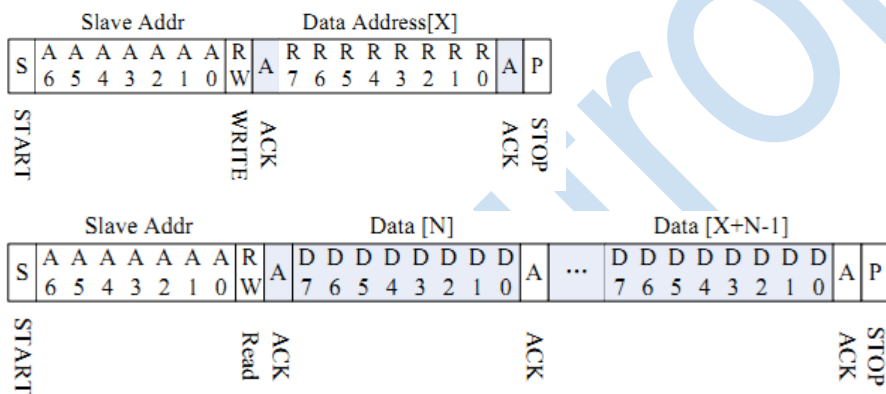
Write multiple bytes to the slave device



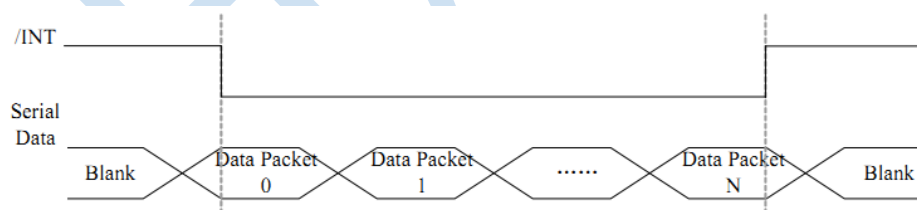
write register address



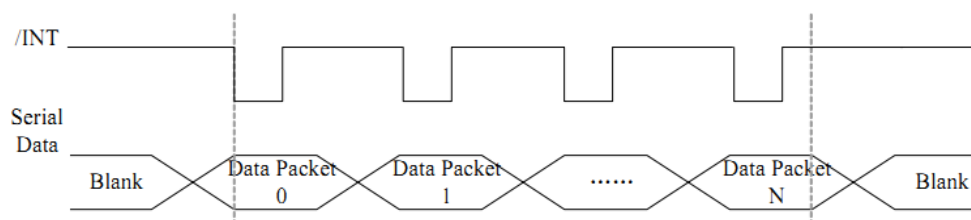
Read multiple bytes from Address[X]



IRQ signal response is divided into two modes: edge-triggered and level-triggered
low level trigger



edge trigger



4. Register Description

The working mode switching command is as follows

Operating mode	command	describe
NOMAL	BE00	Normal reporting and gesture reporting
DBG_IDAC	F001	Factory test data
DBG_POS	BE01	Factory test button and coordinate
DBG_RAW	BF06	Get the rawdata
DBG_SIG	BF07	get the signal

After the IC is powered on, it works in NOMAL mode by default.

NOMAL register description

On-chip Register description											
Address	Name	bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0	Description	Access
00h	Reserver										R/W
01h	Prox_state	Prox_state[7:0]								default: 00 far away: E0 near: C0	R
02h	touch num					touch points[3:0]					R
03h	touch1_XH	event_flg				X_position[11:8]				touch1 event_flg: press: 00 holde: 02 release: 01	R
04h	touch1_XL	X_position[7:0]									R
05h	touch1_YH	touch_ID[3:0]				Y_position[11:8]				touch_ID: finger id	R
06h	touch1_YL	Y_position[7:0]									R
07h	touch1_pres									pressure default: 00	R
08h	Touch1_area									Touch area default: 00	R
9h	touch2_XH	event_flg				X_position[11:8]				touch2 event_flg: press: 00 holde: 02 release: 01	R
10h	touch2_XL	X_position[7:0]									R
11h	touch2_YH	touch_ID[3:0]				Y_position[11:8]				touch_ID: finger id	R
12h	touch2_YL	Y_position[7:0]									R
13h	Touch2_pres									pressure default: 00	R
14h	Touch2_area									Touch area default: 00	R
...	...										
A5h	Sleep_mode	Deepsleep[7:0]								write 03 into Deepsleep	W
A6h	Fw_version	Fw_version[7:0]								Firmware version	R

A7h	Fw_version	Fw_version[15:8]								number	R
A8h	Module_ID	Module_version[7:0]								Mod ID	R
A9h	Project_name	Project_name[7:0]								default: 00	R
AAh	Chip_type	Chip_type[7:0]								Chip model	R
ABh	Chip_type	Chip_type[15:8]									R
ACh	Checksum	Checksum[7:0]								Firmware Checksum	R
ADh	Checksum	Checksum[15:8]									R
...	...										
B0h	Prox_mode	Prox_mode[7:0]								Write 01H Enter Proximity mode Write 00H Exit Proximity mode	W
...	...										
D0h	Ges_mode	Ges_mode[7:0]								Write 01H enter gesture mode Write 00H exit gesture mode	W
...	...										
D3h	Gesture ID	Gesture[7:0]								when gesture mode is enabled double klick:0x24 up:0x22 down:0x23 left:0x20 right:0x21 C:0x34 e:0x33 m:0x32 O:0x30 S:0x46 V:0x54 W:0x31 Z:0x65	R
D4h DAh	Gesture data									make compatible reservations	R
											R
											R

Read version information:

```

write_buf[0] = 0xA1;
hyn_wr_bytes(write_buf, 1, buf_read, 10);
ic->fw_project_id = (buf_read[0] << 24) | (buf_read[1] << 16) | (buf_read[2] << 8) | buf_read[3];
ic->fw_chip_type = buf_read[9];
ic->fw_ver = buf_read[5];
ic->fw_module_id = buf_read[7];

```

Read some coordinate information:

```
write_buf[0] = 0x02;
hyn_wr_bytes(write_buf,1,read_buf,(3+6*2));
rp_buf.rep_num = read_buf[0]&0x0F;
for(i = 0 ; i < 2 ; i++){
    id = (read_buf[3 + i*6] & 0xf0)>>4;
    event = (read_buf[1 + i*6]&0xC0)>>6; //press:0 hold:2 release:1 erro:3
    if(id > 1 || event==3) continue;
    rp_buf.pos_info[index].pos_id = id;
    rp_buf.pos_info[index].event = 1==event ? 0:1;
    rp_buf.pos_info[index].pos_x = ((u16)(read_buf[1 + i*6] & 0x0f)<<8) + read_buf[2 + i*6];
    rp_buf.pos_info[index].pos_y = ((u16)(read_buf[3 + i*6] & 0x0f)<<8) + read_buf[4 + i*6];
    index++;
}
```

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